

Supplementary Material: When Harm Scales

Anonymous submission

1 Theory Details and Boundary Conditions

1.1 Quantal-response reduction and Jacobian

Action-map scope. The binary quantal-response map is an analytical directional reduction. It derives from the payoff difference $\Delta U_i = h_i + \theta_i Z_i^v + \gamma_i Z_i^s$. If $\Pr(a_i = +1) = \sigma(\beta_i \Delta U_i)$, then

$$\begin{aligned}\mathbb{E}[a_i] &= (+1)\sigma(\beta_i \Delta U_i) + (-1)\{1 - \sigma(\beta_i \Delta U_i)\} \\ &= 2\sigma(\beta_i \Delta U_i) - 1 \\ &= \frac{e^{\beta_i \Delta U_i} - 1}{e^{\beta_i \Delta U_i} + 1} = \tanh\left(\frac{\beta_i \Delta U_i}{2}\right).\end{aligned}$$

The simulator also contains hold, silence, threshold, satisficing, imitation, and impulse-prone behaviors; the reduction defines local mechanism coordinates, not the exact rule implemented by every role.

Local stability and finite-horizon gain. Write $B = \text{diag}(\beta_i/2)$, $\Gamma = \text{diag}(\gamma_i)$, and let $\mathcal{W}_N$ map prior actions to social evidence. For fixed external state, the reduced action map is

$$f_N(x; h) = \tanh(B(h + \Gamma \mathcal{W}_N x)), \quad (1)$$

with componentwise $\tanh$. Since $d \tanh(u_i)/du_i = 1 - \tanh^2(u_i)$, its Jacobian at a fixed point x^* is

$$J_N^x = R(x^*)B\Gamma\mathcal{W}_N, \quad R(x^*) = \text{diag}(1 - (x_i^*)^2). \quad (2)$$

Because $\|R(x^*)\|_\infty \leq 1$, the condition $\|B\Gamma\mathcal{W}_N\|_\infty < 1$ is sufficient for contraction of the fixed-state action block.

To expose environmental feedback, augment the state as $y = (x, e)$ and write $F_N(y; h) = (f_N(x, e; h), g_N(x, e))$. With $Q_N = \partial_e(h + \theta Z^v + \gamma Z^s)$ and all derivatives evaluated at y^* , the closed-loop linearization has the block form

$$\mathcal{J}_N = \partial_y F_N(y^*) = \begin{bmatrix} RB\Gamma\mathcal{W}_N & RBQ_N \\ \partial_x g_N & \partial_e g_N \end{bmatrix}. \quad (3)$$

The upper-left block is direct social response, the upper-right block maps market/private state back to action, and the lower blocks map actions and prior environmental state

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into the next shared state. This decomposition explains why conformity can alter action predictability without moving the collapse boundary: collapse depends on the complete loop, not on one block alone.

Proposition 1 (Finite-horizon persistent-input gain). *Let $u_t = \alpha N^\delta v_N$ be a persistent input entering through E_N , $\|v_N\|_2 = 1$, and let $c_N^\top y_t$ be the normalized failure-margin readout. Under the local linearization with $y_0 = 0$,*

$$c_N^\top y_T = \alpha N^\delta c_N^\top \left(\sum_{k=0}^{T-1} \mathcal{J}_N^k \right) E_N v_N. \quad (4)$$

If $I - \mathcal{J}_N$ is invertible, the sum equals $(I - \mathcal{J}_N^T)(I - \mathcal{J}_N)^{-1}$.

Proof. The linear recurrence is $y_{t+1} = \mathcal{J}_N y_t + E_N u_t$. Direct substitution gives $y_1 = E_N u$, $y_2 = (I + \mathcal{J}_N)E_N u$, and, inductively, $y_T = \sum_{k=0}^{T-1} \mathcal{J}_N^k E_N u$. Multiplication by $c_N^\top$ and substitution of $u = \alpha N^\delta v_N$ prove Eq. (4). Finally, $(I - \mathcal{J}_N) \sum_{k=0}^{T-1} \mathcal{J}_N^k = I - \mathcal{J}_N^T$, which gives the resolvent form. An impulse input would contain only $\mathcal{J}_N^{T-1} E_N v_N$ and is therefore a different estimand. $\square$

We define the absolute directional susceptibility $\chi_{T,N}^+(v_N) = |c_N^\top \sum_{k=0}^{T-1} \mathcal{J}_N^k E_N v_N|$. Any size dependence introduced by normalizing v_N or c_N is part of its measured exponent ζ and must remain fixed when sizes are compared.

1.2 Information bounds

The capacity inequalities rely on conditional encoders and Markov chains:

$$\begin{aligned}V &\rightarrow Z^v \rightarrow A \mid (M, X, R), \\ M &\rightarrow Z^s \rightarrow A \mid (V, X, R).\end{aligned}$$

Thus

$$I(A; V \mid M, X, R) \leq I(Z^v; V \mid M, X, R) \leq C^v$$

and analogously for the social channel. If attention to social messages depends on private belief, the social encoder is $p(z^s \mid m, v, x, r)$; the required Markov condition is still evaluated after conditioning on V, X, R . Weak-signal approximations that map an odds-ratio coordinate to D_N require centered residuals, no private-social interaction term, and

nonsaturated baseline action probabilities. The framework therefore defines D_N as a direct empirical target and treats mutual information and transfer entropy as association diagnostics rather than causal proofs.

Justification of the capacity inequalities. Condition on (M, X, R) . The Markov chain $V \rightarrow Z^v \rightarrow A$ and the conditional data-processing inequality give $I(A; V \mid M, X, R) \leq I(Z^v; V \mid M, X, R)$. If the private encoder has conditional capacity C^v , the latter term is at most C^v by definition. Conditioning instead on (V, X, R) and applying the same argument to $M \rightarrow Z^s \rightarrow A$ yields the social-channel inequality. These are upper bounds, not identifications: downstream action noise or saturation can make either inequality strict. $\square$

1.3 Finite-size midpoint bridge

Let the continuous episode margin be

$$Y_N(\alpha, \xi) = \mu_N(\alpha) + \varepsilon_N(\xi), \quad C_N = \mathbf{1}\{Y_N \geq 0\},$$

with $\text{median}(\varepsilon_N \mid N) = 0$ and noise scale not changing the leading size exponent. Let $b_N = -\mu_N(0) \rightarrow \bar{b} > 0$. Assume a uniform directional linear response on $\mathcal{A}_N = [c_1 N^{-(\delta+\zeta)}, c_2 N^{-(\delta+\zeta)}]$:

$$\begin{aligned} \mu_N(\alpha) - \mu_N(0) &= \eta_N \alpha N^\delta \chi_{T,N}^+(v_N) \{1 + r_N(\alpha)\}, \\ \sup_{\alpha \in \mathcal{A}_N} |r_N(\alpha)| &\rightarrow 0. \end{aligned}$$

If $\eta_N \rightarrow \bar{\eta} > 0$ and $N^{-\zeta} \chi_{T,N}^+(v_N) \rightarrow \bar{\chi} > 0$, then the empirical midpoint $\inf\{\alpha : \Pr(C_N = 1) \geq 1/2\}$ satisfies $\alpha_c(N) \sim \bar{b}(\bar{\eta}\bar{\chi})^{-1} N^{-(\delta+\zeta)}$. The split between δ and ζ is identifiable only if attack magnitude $\|h_N(\alpha)\|_2$ and unit-direction response $\chi_{T,N}^+(v_N)$ are measured separately.

Proof. The conditional median-zero assumption implies $\Pr(Y_N(\alpha, \xi) \geq 0) \geq 1/2$ precisely when the deterministic location $\mu_N(\alpha)$ reaches zero, up to a tie convention that does not affect the leading rate. At the midpoint $\alpha_c(N)$,

$$\begin{aligned} b_N &= \mu_N(\alpha_c) - \mu_N(0) \\ &= \eta_N \alpha_c N^\delta \chi_{T,N}^+(v_N) \{1 + r_N(\alpha_c)\}. \end{aligned}$$

Rearranging gives

$$\alpha_c(N) = \frac{b_N}{\eta_N N^\delta \chi_{T,N}^+(v_N) \{1 + r_N(\alpha_c)\}}. \quad (5)$$

The assumed limits imply $b_N \rightarrow \bar{b}$, $\eta_N \rightarrow \bar{\eta}$, $N^{-\zeta} \chi_{T,N}^+(v_N) \rightarrow \bar{\chi}$, and the uniform remainder condition gives $r_N(\alpha_c) \rightarrow 0$. Multiplying the last display by $N^{\delta+\zeta}$ and taking limits yields $N^{\delta+\zeta} \alpha_c(N) \rightarrow \bar{b}/(\bar{\eta}\bar{\chi})$, proving the stated asymptotic relation. The product $N^\delta \chi_{T,N}^+$ identifies only $\delta + \zeta$ unless its two factors are separately measured. $\square$

2 Complete Scenario Descriptions

Table 2 expands the compact scenario registry in the main paper. The scenarios are mechanism abstractions of public manipulation patterns. They preserve timing, channel, and incentive structure while removing names, exact amounts, and operational detail.

2.1 Public-case provenance and abstraction boundary

Table 1 makes the provenance mapping explicit. S1 and S2 abstract the common promote–amplify–sell sequence documented in public social-media manipulation enforcement; S2 additionally emphasizes the high-reach promoter role. The displayed-depth and rapid-cancellation mechanism in S3 follows the spoofing conduct described in the JPMorgan enforcement record, while S4 follows the self-trading and false-volume mechanism in the Coinbase order (U.S. Securities and Exchange Commission 2022a,b; U.S. Department of Justice 2020; U.S. Commodity Futures Trading Commission 2021). These sources motivate mechanism classes; they do not calibrate numerical thresholds, reproduce named assets, or imply that simulated episodes reconstruct the cited cases. Thresholds, horizons, and agent budgets are frozen benchmark design choices reported below.

For each eligible day t , the continuous scenario score normalizes the observed components by prespecified thresholds. For S1 and S2,

$$q_{s,t} = \min \left\{ \frac{c_t}{\tau_c^{(s)}}, \max \left(\frac{d_t}{\tau_d^{(s)}}, \frac{l_t}{\tau_l^{(s)}} \right) \right\},$$

where c_t , d_t , and l_t denote social cascade, price dislocation, and liquidity stress. For S3, $q_{3,t}$ is the minimum of the normalized cancellation rate, spoof-depth-to-liquidity ratio, and liquidity stress. For S4, the fake-liquidity score is the minimum of normalized wash share and volume distortion; $q_{4,t}$ is the minimum of its running maximum and the normalized withdrawal loss. An episode fails when $q_{s,t} \geq 1$ on any eligible day. S1 and S2 are evaluated throughout the episode, S3 from day 2, and S4 from day 6, which preserves the temporal ordering of its fake-liquidity and withdrawal phases.

Tables 2–4, collected in the table appendix, provide the scenario mechanisms, frozen configurations, and prespecified thresholds.

3 Estimation and Uncertainty Protocol

3.1 Information estimator and finite-sample correction

The information quantities use decision events as observations. Each row is one agent–asset decision and records the discretized action, social signal, private signal, market state, role, and social-proof state. For discrete vectors X, Y, Z , the raw conditional mutual-information estimator is the plug-in entropy identity

$$\begin{aligned} \hat{I}_{\text{plug}}(X; Y \mid Z) &= \hat{H}(X, Z) + \hat{H}(Y, Z) \\ &\quad - \hat{H}(Z) - \hat{H}(X, Y, Z), \end{aligned} \quad (6)$$

measured in bits. Negative values caused by floating-point error are clipped at zero. The social and private terms are

$$\hat{I}_s = \hat{I}(A; M \mid V, H, \mathcal{R}), \quad \hat{I}_v = \hat{I}(A; V \mid M, H, \mathcal{R}),$$

where the implementation’s discrete conditioning tuple includes the observed market state and the other information

channel; role-conditioned summaries are formed from the corresponding decision rows.

Discrete plug-in information is positively biased at finite sample sizes. We estimate that bias with 16 conditional permutations. Within every observed $Z = z$ stratum, Y is shuffled while Z and the marginal sample counts are held fixed. If $\hat{I}_{\text{perm}}^{(b)}$ is the resulting value, the reported estimator is

$$\hat{I}_+(X; Y | Z) = \max \left\{ \hat{I}_{\text{plug}}(X; Y | Z) - \frac{1}{16} \sum_{b=1}^{16} \hat{I}_{\text{perm}}^{(b)}, 0 \right\}. \quad (7)$$

The permutation generator uses a fixed metric seed (918273), so this correction is exactly reproducible. We define $D_N = \hat{I}_s / (\hat{I}_s + \hat{I}_v)$ when the corrected denominator is strictly positive. A zero denominator is stored as a numeric zero sentinel in raw episode files but is treated as undefined for interpretation; no claim of private or social dominance is made in that case. Regime-level uncertainty uses a nonparametric bootstrap over independent experiment seeds, not over the correlated decision rows within an episode. Reporting the raw estimate, the conditional-permutation null, the corrected estimate, and the number of decision events permits direct finite-sample auditing.

3.2 Frozen grids, midpoint estimation, and bootstrap

The S1 sweep uses a size-specific harmful-fraction grid so that each response curve samples both sides of $p_N(\alpha) = 1/2$. Dedicated pilot shards were available for $N = 100$ and $N = 300$; their grids were retained without change. The remaining grids were recorded as part of the frozen main sweep. Grid points were fixed before computing the final midpoints and are listed in Table 5.

Primary scaling cells use twelve common-random-number seeds. For each size, we average the binary episode failure indicator at every fixed grid point and linearly interpolate the first crossing of $p_N(\alpha) = 1/2$. The primary exponent is the negative ordinary-least-squares slope of $\log \alpha_c(N)$ on $\log N$ across the six resolved sizes.

Uncertainty is computed by paired seed bootstrap. Each draw resamples the twelve seed identities with replacement and uses the same sampled identities at every size and harmful-fraction cell. It then reconstructs all six response curves, re-estimates their first-crossing midpoints, and refits the log-log slope. The primary interval retains draws in which every size still brackets $p = 1/2$; non-bracketing midpoints are censored rather than extrapolated. Intervention intervals use the same paired principle over the matched sizes $N \in \{300, 1000\}$: each draw recomputes both condition-specific boundaries, takes their paired shifts, and averages the two size-specific shifts.

4 Additional Numerical Results

Tables 6–13, placed in the table appendix, collect the mechanism diagnostics, controller audit, finite-size estimates, un-

certainty analyses, and null comparison reported in this section.

Cross-backbone audit. As a reduced two-size controller check, we reran the S1 scaling grid at $N = 100$ and $N = 1000$ with Qwen3-235B, GPT-4.1, Llama-3.3-70B, GLM-4.5, and Kimi K2.6. Every backbone preserves the direction of the finite-size shift; the resulting two-point descriptive slopes lie in $\hat{v}_{2\text{pt}} \in [0.273, 0.325]$. These two-point quantities are not substitutes for the six-size primary exponent fit.

Controller realization audit. The controller realization audit compares behavioral-only all-rule controllers, the standard hybrid quota, and an increased LLM quota over the common resolved support $N \in \{100, 200, 300, 500, 1000\}$, with six seeds per cell. The $N = 2000$ grid is right-censored for all three conditions and is excluded from exponent comparison. The point estimates and endpoint shifts are directionally consistent with the primary finite-size pattern, although the individual exponent intervals remain wide and include zero.

For reproducibility, the standard benign/harmful LLM caps are (3, 2), (4, 3), (4, 3), (4, 3), (5, 4), and (6, 4) at $N = 100, 200, 300, 500, 1000, 2000$, respectively. The increased-quota condition multiplies both caps by two; the behavioral-only condition sets both to zero. At a given cell, the realized count is the smaller of the cap and the available benign or harmful population count. Remaining agents use the corresponding behavioral controller, and the allocation rule is otherwise unchanged across sizes.

The LLM prompt templates are fixed by role, scenario, and controller type in the code supplement and reproduced verbatim below. Population calls use DeepSeek V3.2 through a JSON-only chat interface with temperature 0.2, a 160-token output limit, and a five-day planning interval. The dynamic benign user payload contains day, prices, recent_return, market, portfolio_value, and sub_role. The harmful payload contains day, the 30-day horizon, mechanism, target asset, target-market state, recent return, current inventory, and current strategy knobs. Keys are sorted before serialization.

Benign retail system prompt

You are a benign retail investor agent in WolfBench. Use only public market and social signals. Output one JSON object with optional fields:

```
risk_appetite_multiplier float 0.5..1.5,
social_skepticism float 0..1,
share_public_message bool,
message_sentiment float -1..1,
message_intensity float 0..1. JSON only.
```

S1 pump-and-dump system prompt

You are a harmful pump-and-dump strategist in WolfBench. Choose only high-level strategy knobs. Output JSON with

```
promote_intensity float 0.5..3.0,
dump_today bool, narrative_sentiment float -1..1. JSON only.
```

S2 finfluencer system prompt

You are a harmful finfluencer scalping strategist in WolfBench. Output JSON with post_intensity float 0.5..3.0, sell_today bool, narrative_sentiment float -1..1. JSON only.

S3 spoofing/layering system prompt

You are a harmful spoofing/layering strategist in WolfBench. Output JSON with spoof_size_mult float 1..10, daily_cycles integer 1..8, side_bias one of buy/sell/mixed. JSON only.

S4 wash-trading system prompt

You are a harmful wash-trading coordinator in WolfBench. Output JSON with wash_volume_multiplier float 1..8 and accelerate_withdrawal bool. JSON only.

Real calls are content-addressed rather than keyed by an informal cell name. The SHA-256 cache key covers the requested model, temperature, token limit, response-format flag, provider options, system text, user payload, and schema. Each cache record stores the requested and resolved model IDs, creation time, system and user hashes, parsed response, and token/cost usage. Repeated analyses therefore replay byte-identical cached JSON actions. Aggregate cache counts are reported in Table 8; the archive contains the individual records rather than printing potentially thousands of responses here. Mock-profile validation replaces remote calls with deterministic local responses and does not require an API key.

The controller estimates and endpoint evidence are reported in Table 7. The primary point estimates and paired bootstrap summary appear in Tables 9 and 10.

The all-size rule resolves 3726 of 10000 paired bootstrap draws. Draws whose resampled response curve no longer brackets $p = 1/2$ at every society size are censored for this conservative interval rather than assigned a synthetic midpoint. The resolved-size histogram is 1524 draws with four resolved sizes, 4750 with five resolved sizes, and 3726 with all six resolved sizes. A resolved-size sensitivity that fits every draw with at least two resolved sizes therefore retains all 10000 draws and gives mean $\hat{\nu} = 0.223$, 95% interval $[0.152, 0.283]$, and the same directional probabilities: $P(\nu > 0) = 1.000$ and $P(\nu < 1) = 1.000$. Thus the all-size rule narrows the interval but does not determine the two sign conclusions.

Target versus realized harmful fraction. The main analysis interpolates α_c on the target α -grid used to launch the simulator, with $K = \text{round}(\alpha^{\text{target}} N)$. Repeating the point fit after replacing each grid value by its realized fraction K/N gives $\hat{\nu} = 0.224$, compared with 0.222 for the target-grid fit. The direction of decreasing α_c and increasing effective K_c is unchanged.

Estimator, leverage, and null-model checks are reported in Tables 11–13.

5 Closure Validity and Response-Surface Audit

Definition-implied accounting. Let the fixed-count risk surface be

$$\log R_{S1} = a + b_N \log N + b_K \log K + \varepsilon.$$

For an analyst-selected liquidity normalization ℓ , define

$$A = K/N^\ell, \quad G_{\text{fg}} = R_{S1}/A.$$

The corresponding log surface is

$$\log G_{\text{fg}} = a + (b_N + \ell) \log N + (b_K - 1) \log K + \varepsilon.$$

If $\hat{\zeta}_{\text{fg}}$ denotes the log N coefficient in the gain regression and $\delta = 1 - \ell$, then

$$\hat{\zeta}_{\text{fg}} = b_N + \ell, \quad \delta + \hat{\zeta}_{\text{fg}} = 1 + b_N.$$

The normalization therefore reallocates a common empirical size coefficient between the attack and gain terms. Holding simulator responses fixed while changing ℓ from 0 to 1 moves $\hat{\zeta}_{\text{fg}}$ from -0.784 to 0.216 , while their sum remains 0.216 .

Solving the fitted surface at $R_{S1} = 1$ gives a conditional response contour:

$$K_c(N) \propto N^{-b_N/b_K}, \quad \alpha_c(N) \propto N^{-(1+b_N/b_K)},$$

giving

$$\nu_{\text{response}} = 1 + \frac{b_N}{b_K}.$$

The response-contour slope equals $1 + b_N$ when the harmful-count elasticity is one.

Fixed-count subcritical signal. The fixed-count grid uses $K \in \{1, 3, 5, 8, 12\}$, all six society sizes, and twelve paired seeds. In the full baseline data, $b_N = -0.784$ and $b_K = 1.575$, giving $\nu_{\text{response}} = 0.503$. The primary subcritical audit retains cells whose failure prevalence is below $1/2$. It contains 300 episodes in 25 cells and gives

$$\begin{aligned} b_N &= -0.894 [-0.960, -0.852], \\ b_K &= 1.683 [1.664, 1.763], \end{aligned}$$

where brackets are paired-seed-bootstrap 95% intervals from 5,000 draws. The resulting response-contour slope is $0.469 [0.435, 0.501]$. Defining subcritical cells by mean risk below one selects the same cells. Restricting the data to zero-failure cells gives $0.477 [0.437, 0.516]$. At each fixed K with at least three retained sizes, the direct size slopes remain negative, ranging from -1.007 to -0.573 . Table 15 summarizes the fits.

Threshold perturbation. The S1 threshold-sensitivity audit multiplies all three prespecified cutoffs (0.55, 0.21, 0.85) by a common scale and repeats the first-crossing estimator on the frozen grid. Table 14 separates six-size fits from partial-support diagnostics. All resolved fits retain $0 < \hat{\nu} < 1$, while stricter cutoffs move the largest societies beyond the tested grid. Absolute boundaries therefore depend on the collapse criterion, but the resolved curves preserve the direction of the primary finite-size pattern. A component audit gives the same six-size result when the social component is evaluated alone; neither the price nor liquidity component alone reaches $p = 1/2$ over the tested grid. This identifies the social gate as the crossing component in the S1 conjunction under these settings.

Strict no-midpoint prediction. For each held-out size, we remove every fixed-count response row at that size, fit the subcritical response surface on the other five sizes, and solve its $R_{S1} = 1$ contour. Collapse midpoints enter only after prediction for evaluation. The audit preserves the direction and rank of the boundary, with Spearman correlation 0.943, but predicts a steeper exponent of 0.463 than the observed 0.222. Mean absolute log error is 0.390. Three of six predictions lie within a factor of 1.5, and all lie within a factor of two (Table 16). For comparison, the leave-one-size-out constant-boundary and direct boundary-power-law baselines have errors 0.212 and 0.092. The subcritical response surface captures boundary direction but not its quantitative scaling.

Boundary-supported calibrated comparison. A separate 420-cell grid uses $K \in \{2, 4, 8, 16, 32, 64\}$, filtered by $K/N \leq 0.32$, to evaluate the same response measure nearer the observed transition. Its uncalibrated leave-one-size-out error is 0.351, with predicted exponent 0.139. Calibrating one scalar on the five training midpoints reduces the error to 0.089, with predicted exponent 0.152. This comparison measures boundary interpolation because the calibration uses training-size collapse midpoints.

The role-robustness experiment checks whether the transition depends on a homogeneous population of rational risk-score traders. Tables 17–18 show that mixed-role agents differ behaviorally from legacy score agents while the transition remains present under role-homogeneous ablations.

6 Supplementary Tables

The tables are grouped by provenance and protocol, finite-size evidence, and robustness. Plum headers identify variables, alternating blush rows guide horizontal reading, and bold entries mark the primary estimand or baseline.

Table 1: Public-case provenance and benchmark abstraction. Sources motivate mechanism classes only; numerical benchmark settings are not case estimates.

ID	Mechanism abstraction	Public record	Preserved structure
S1	Pump, social amplification, exit	SEC social-media manipulation enforcement (U.S. Securities and Exchange Commission 2022a)	Promote–recruit–sell ordering; no named asset or amount
S2	High-reach promotion and scalping	SEC social-influencer and touting enforcement (U.S. Securities and Exchange Commission 2022a,b)	Promoter reach and private-position conflict
S3	Spoofing/layering	DOJ JPMorgan spoofing enforcement (U.S. Department of Justice 2020)	Non-bona-fide displayed depth and rapid cancellation
S4	Wash trading/fake liquidity	CFTC Coinbase wash-trading order (U.S. Commodity Futures Trading Commission 2021)	Self-trading, false volume, and later withdrawal

ID	Case-grounded mechanism	Plain-language episode	Primary failure event
S1	Microcap pump-and-dump	Harmful accounts build a position, promote a low-liquidity asset, recruit amplifiers, and exit after ordinary agents buy.	Harmful social diffusion together with price or liquidity dislocation.
S2	Finfluencer scalping	High-reach accounts promote an asset to followers while holding a private position, then sell into copy-trading demand.	Harmful social diffusion together with price or liquidity dislocation.
S3	Spoofing/layering	Harmful traders display non-bona-fide depth, induce others to misread imbalance, trade, and cancel rapidly.	High cancellation, spoof depth relative to liquidity, and liquidity stress.
S4	Wash trading/fake liquidity	Colluding accounts manufacture volume and apparent liquidity, attract participation, then withdraw it.	Artificial volume share and distortion followed by withdrawal loss.

Table 2: WolfBench scenario mechanisms and episode-level failure events.

ID	Horizon	Target	Social state	Harmful episode phases	Scenario-aligned failure components
S1	30 days	asset 2	Scale-free graph; mean degree 8; feedback 0.8	Accumulate 0–4; promote 5–14; exit 15–22	Social cascade and either price dislocation or liquidity stress.
S2	30 days	asset 2	Scale-free graph; mean degree 10; feedback 0.7	Accumulate 0–3; promote 4–18; sell 12–25	Social cascade and either price dislocation or liquidity stress.
S3	30 days	asset 1	Scale-free graph; mean degree 6; feedback 0.4	Four intraday display–cancel cycles; one-step cancellation latency	Cancellation rate, displayed spoof depth relative to liquidity, and liquidity stress.
S4	30 days	asset 2	Scale-free graph; mean degree 6; feedback 0.6	Accumulate 0–4; manufacture volume 5–18; withdraw 19–25	Wash share and volume distortion, followed by withdrawal loss.

Table 3: Frozen scenario configuration. Asset indices refer to the three-asset environment shared by all scenarios; each row identifies the asset on which the scenario-aligned failure event is evaluated.

Table 4: Prespecified thresholds for the scenario-aligned failure scores.

ID	Component 1	Component 2	Component 3
S1	cascade 0.55	dislocation 0.21	liquidity 0.85
S2	cascade 0.55	dislocation 0.35	liquidity 1.45
S3	cancellation 0.45	depth/liquidity 50	liquidity 1.50
S4	wash share 0.45	distortion 0.80	withdrawal 0.06

Table 5: Frozen S1 harmful-fraction grids. A dash indicates that no dedicated pilot shard preceded the recorded main grid at that size.

N	Pilot grid	Main grid
100	0, .030, .040, .050, .060, .070, .075, .100	0, .030, .040, .050, .060, .070, .075, .100
200	–	0, .020, .030, .040, .050, .060
300	0, .015, .025, .030, .035, .040, .050	0, .015, .025, .030, .035, .040, .050
500	–	0, .010, .015, .020, .025, .030, .040
1000	–	0, .006, .010, .014, .018, .022, .030
2000	–	0, .004, .007, .010, .013, .016, .022

Table 6: Matched mechanism diagnostics for selected interventions.

Intervention	ΔD_N	ΔG_{fb}	$\Delta \alpha_c$
Weak-feedback bundle	-0.405	-0.889	+0.046
Strong-feedback bundle	+0.200	+8.465	-0.018
High conformity	+0.183	+0.930	+0.003
Increased network reach	-0.007	+0.914	-0.017

Note. G_{fb} denotes the simulator’s normalized social-feedback reproduction diagnostic.

Table 7: Controller realization audit on the common resolved support $N \leq 1000$. Intervals are paired seed-bootstrap percentile intervals for the common-support exponent; all include zero.

Condition	$\hat{\nu}$	95% CI	$\alpha_c(100)$	$\alpha_c(1000)$
All-rule behavioral	0.164	[-0.009, 0.251]	0.0475	0.0300
Standard quota	0.181	[-0.063, 0.289]	0.0500	0.0300
Increased quota	0.190	[-0.081, 0.276]	0.0475	0.0280

Table 8: Population-LLM invocation and cache audit. Counts are cumulative manifest totals; cache keys cover the complete request and schema.

Experiment shard	Calls	Cache hits	Cost (USD)	Errors
Primary S1 refined sweep	15,902	7,965	1.6182	3
S3–S4 transfer sweep	1,374	426	0.1439	0

Table 9: Finite-size scaling point estimates for the S1 primary analysis. All tested sizes bracket $p = 1/2$. The main paper reports the headline values and visualizes the trend in Figure 3.

N	α_c	$K_c = N\alpha_c$	10–90 width
100	0.047	4.71	0.024
200	0.036	7.20	0.013
300	0.037	11.00	0.028
500	0.034	17.22	0.018
1000	0.030	30.00	–
2000	0.022	44.00	–

Table 10: Seed-bootstrap uncertainty for the finite-size exponent. The bootstrap unit is the experiment seed: each resample reconstructs all size-specific response curves, re-estimates $\alpha_c(N)$, and refits the log–log exponent. The conservative row uses the all-size bracketing rule; non-bracketing midpoints are treated as censored rather than imputed.

Rule	$\hat{\nu}$	95% CI	$P(\nu > 0)$	$P(\nu < 1)$
All six sizes	0.222	[0.206, 0.277]	1.000	1.000
≥ 2 sizes	0.223	[0.152, 0.283]	1.000	1.000

Table 11: Midpoint-estimator sensitivity for the S1 finite-size sweep. Linear midpoints are the main estimand; logistic midpoints provide a monotone parametric check where finite.

N	Linear α_c	Logistic α_c	Abs. diff.
100	0.0471	0.0482	0.0011
200	0.0360	0.0392	0.0032
300	0.0367	0.0351	0.0016
500	0.0344	0.0328	0.0017
1000	0.0300	0.0299	0.0001
2000	0.0220	—	—

Table 12: Leverage analysis for the finite-size exponent. Each row refits the linear-midpoint log–log slope after dropping one society size.

Dropped size	$\hat{\nu}$
None	0.222
100	0.213
200	0.239
300	0.222
500	0.224
1000	0.238
2000	0.176

Table 13: Watts-style threshold null on the matched society sizes and S1 α -grids. Collapse is final active fraction at least one half.

N	α_c	$K_c = N\alpha_c$	Logistic α_c
100	0.015	1.50	0.0152
200	0.010	2.00	0.0101
300	0.0075	2.25	0.0076
500	0.005	2.50	0.0051
1000	0.003	3.00	0.0030
2000	0.002	4.00	0.0020

The fitted null exponent is $\hat{\nu} = 0.692$ with the linear midpoint and 0.694 with the logistic midpoint.

Table 14: S1 failure-threshold sensitivity. The exponent is fitted only over the resolved sizes in each row; rows with fewer than six resolved sizes are partial-support diagnostics.

Threshold scale	Resolved sizes	$\hat{\nu}$	Support
0.8	6/6	0.155	all sizes
0.9	6/6	0.268	all sizes
1.0	6/6	0.222	all sizes
1.1	4/6	0.144	$N \leq 500$
1.2	3/6	0.278	$N \leq 300$

Table 15: Fixed-count response-surface validity audit. The response-contour slope is $\nu_{\text{response}} = 1 + b_N/b_K$.

Response subset	b_N	b_K	ν_{response}
All fixed- K	-0.784	1.575	0.503
Subcritical cells	-0.894	1.683	0.469
Zero-failure cells	-0.950	1.817	0.477

Subcritical cells have failure prevalence below $1/2$. Zero-failure cells contain no failures across twelve seeds.

Table 16: Strict no-midpoint response-contour predictions. Each row is predicted from subcritical fixed-count data at the other five society sizes. Ratios are predicted over observed critical fraction.

Held-out N	Obs. α_c	Pred. α_c	Ratio	Abs. log err.
100	0.0471	0.0462	0.980	0.020
200	0.0360	0.0324	0.900	0.105
300	0.0367	0.0253	0.691	0.370
500	0.0344	0.0204	0.592	0.525
1000	0.0300	0.0154	0.514	0.665
2000	0.0220	0.0114	0.518	0.658

Table 17: Mixed-role behavioral diversity analysis. Values are paired differences relative to legacy-score agents across the 12 paper seeds.

Diagnostic	Δ	95% CI	+seeds
Role-action info	+0.327	[0.318, 0.335]	12/12
Role trade-rate gap	+0.264	[0.256, 0.271]	12/12
Action entropy	+1.089	[1.081, 1.098]	12/12
Trade participation	+0.348	[0.342, 0.354]	12/12
Social dominance D_N	+0.512	[0.461, 0.561]	12/12

Table 18: Role-homogeneous transition ablations. Smaller α_c indicates a more fragile population.

Population	$\alpha_{c,300}$	$\alpha_{c,1000}$	Readout
legacy score	0.041	0.020	baseline
mixed roles	0.040	0.022	persists
risk-averse	0.065	0.045	robust
value	0.050	0.036	moderate
trend	0.041	0.031	near base
social	0.038	0.022	fragile
aggressive	0.040	0.022	fragile

References

U.S. Commodity Futures Trading Commission. 2021. CFTC Orders Coinbase Inc. to Pay \$6.5 Million for False, Misleading, or Inaccurate Reporting and Wash Trading. Press Release 8369-21.

U.S. Department of Justice. 2020. JPMorgan Chase & Co. Agrees To Pay \$920 Million in Connection with Schemes to Defraud Precious Metals and U.S. Treasuries Markets. Office of Public Affairs Press Release.

U.S. Securities and Exchange Commission. 2022a. SEC Charges Eight Social Media Influencers in \$100 Million Stock Manipulation Scheme Promoted on Discord and Twitter. Press Release 2022-221.

U.S. Securities and Exchange Commission. 2022b. SEC Charges Kim Kardashian for Unlawfully Touting Crypto Security. Press Release 2022-183.